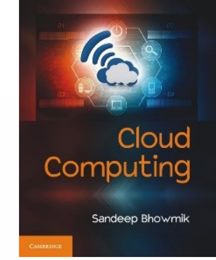
Cloud Computing

Chapter 6

Security Reference Model

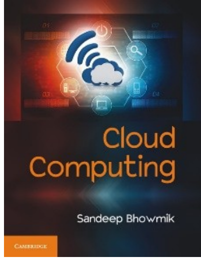
Cambridge University Press



The Security Concern

- Security is one among the topmost concerns of any computing model and cloud computing is no exception.

- In cloud computing, consumers are moving from the traditional in-house computing environment to outside service providers.

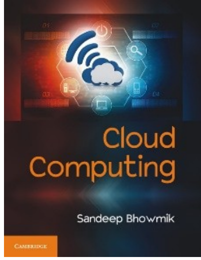


The Security Concern

- Traditional data centers allows perimeterised (i.e. within organization’s own network boundary or perimeter) access to computing resources.

- Cloud computing promotes de-perimeterisation.

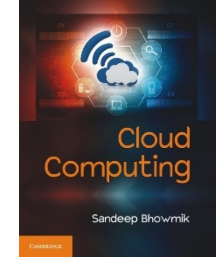
- Traditional concept of security boundary no more applies in cloud computing.



The Security Concern

- Cloud computing moves beyond the concept of working inside protected network boundary.

- But it causes no more threat to security which was not there in traditional computing.

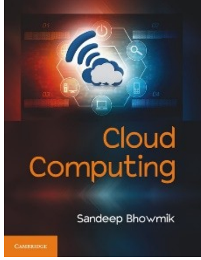


The Security Concern

- **Cloud Security Working Groups**

- Many organizations and groups have worked separately on developing a cloud security model.

- Two such bodies are –
  - The Cloud Security Alliance
  - Jericho Forum Group

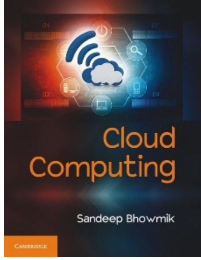


The Security Concern

- **The Cloud Security Alliance (CSA)**

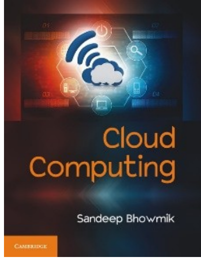
- "Security Guidance for Critical Areas of Focus in Cloud Computing" released by CSA in 2009 is considered as vital testimonial on cloud computing security.

- It categorizes the cloud security related issues in fourteen different sections.



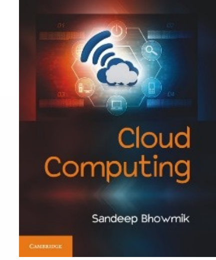
The Security Concern

- **The Jericho Forum Group**
- An international consortium formed with the objective of addressing concerns related to de-perimeterised computing environment.
- They have contributed positively in development of cloud security framework.
- Jericho Forum and the Cloud Security Alliance had worked together to promote best practices for secured collaboration in the cloud.



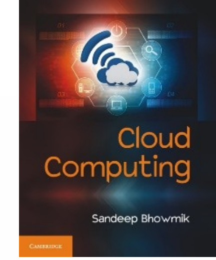
Elements of Cloud Security Model

- Analyst firm Gartner advices consumers to seek transparency related to seven specific issues from service providers before moving into cloud –
  - Privileged user access
  - Regulatory compliance
  - Data location
  - Data segregation
  - Recovery
  - Investigative support
  - Long-term viability



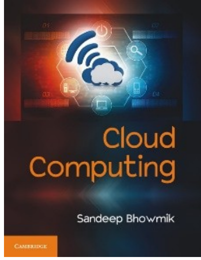
Cloud Security Reference Model

- The cloud computing community and many organizations working in the field of network security were working for years to develop a model to address cloud security.
- The Jericho Forum group came up with a model called *Cloud Cube Model*, to address the security issue.
- This cube model is considered as the security reference model for cloud computing.



The Cloud Cube Model

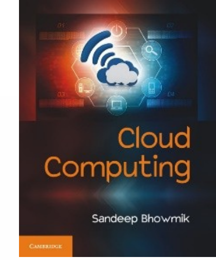
- The Jericho Forum Group proposed Cloud Cube Model in 2009, defining a three-dimensional cube.
- The model was originally created to address the issue of network de-perimeterisation.
- The model suggests that cloud security should not measured depending only on the narrow perspective of ‘internal’ or ‘external’.
- Many other factors are also related with the issue of security.



The Cloud Cube Model

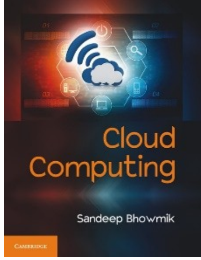
- The cloud cube model is designed to represent four security related criterions.

- Jericho Forum suggests to decide about the four criterions while moving to cloud computing environment –
    - Data Boundary
    - Ownership
    - Security Boundary
    - Sourcing



The Cloud Cube Model

- These four criterions are represented across different dimensions of a cube.
- Each of these 4 criterions have 2 probable answers.
- Hence, there can be 4^2 or 16 different forms of cloud computing environment.

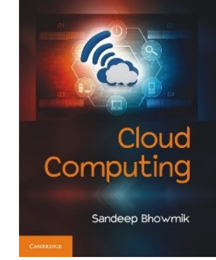


The Cloud Cube Model

- **Data Boundary**
  - **Internal (I)**
  - **External (E)**

- This security dimension represents the physical storage location of organization's data.

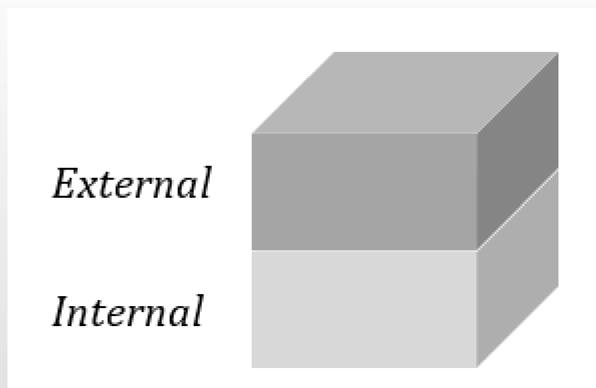
- It is important to note that, external storage location does not necessarily mean lesser security.

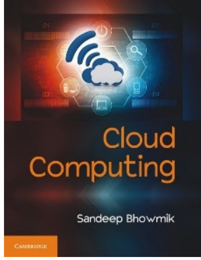


The Cloud Cube Model

- **Data Boundary**

- The *Data Boundary* dimension divides the entire cube of the Cloud Cube model, in two parts.





The Cloud Cube Model

- **Ownership**
  - **Proprietary (P)**
  - **Open (O)**

- This dimension determines the ownership of the technology used for building the cloud.

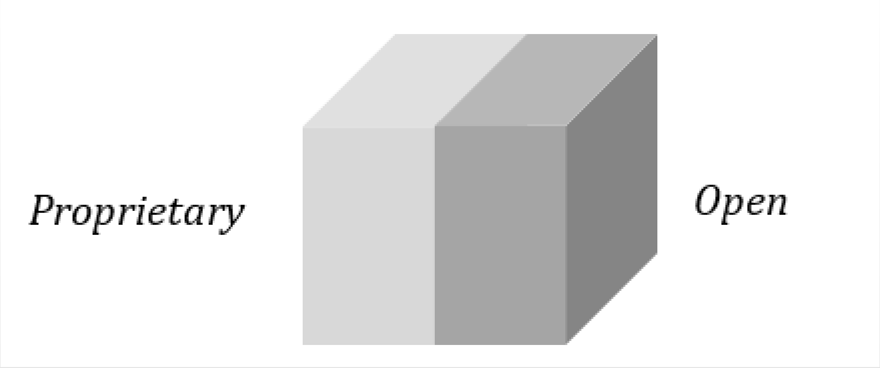
- Reputed commercial vendor generally prefer to build services using their own proprietary technologies.

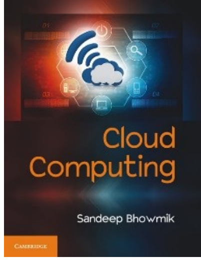
- But, this limits interoperability.

The Cloud Cube Model

- **Ownership**

- The *Ownership* dimension divides the entire cube of the Cloud Cube model, in two parts.





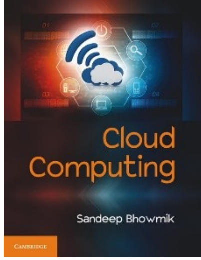
The Cloud Cube Model

- **Security Boundary**
  - **Perimeterised (Per)**
  - **De-perimeterised (D-p)**

- This dimension determines whether to operate inside the traditional network security boundary or not.

- Perimeterised approach enhances security, but prevents collaboration.

- De-perimeterised system shows the natural intent to collaborate with the systems outside own perimeter.



The Cloud Cube Model

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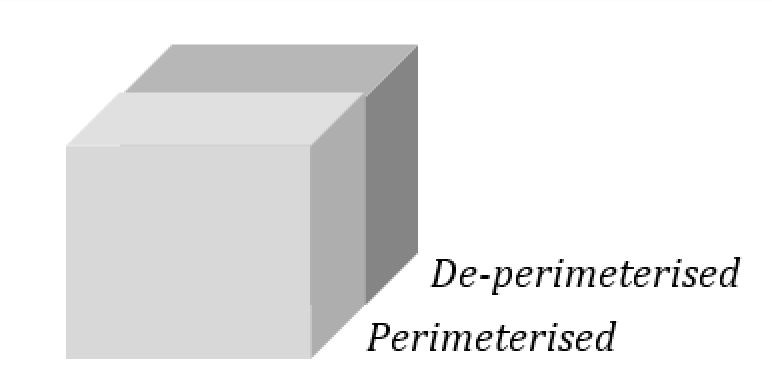
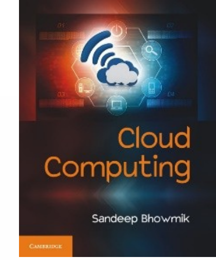


The Cloud Cube Model

- Depending on data-boundary (I/E) and ownership (P/O) there can be four types of cloud formations - IP, IO, EP, EO.

- Each of these forms comes with either of the two architectural mindsets - Perimeterised or De-perimeterised, as security boundary.

- Taken together, there are total eight possible cloud formations - Per (IP, IO, EP, EO) and D-p (IP, IO, EP, EO).



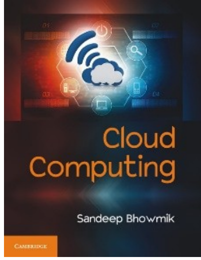
The Cloud Cube Model

- **Sourcing**
  - **Insourced**
  - **Outsourced**

- This security dimension indicates who delivers and manages the service.

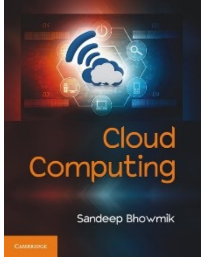
- If it is provided by organization’s own/internal team, then it is *Insourced*.

- If the service is delivered by some third party then it is called *Outsourced*.



The Cloud Cube Model

- **Sourcing**
- Insourced cloud services indicates towards private cloud.
- Outsourced service can deliver both public and private cloud.
- Insourcing of service does not mean better security. Security of cloud service largely depends on the expertise of the delivery team.



The Cloud Cube Model

- **Sourcing**

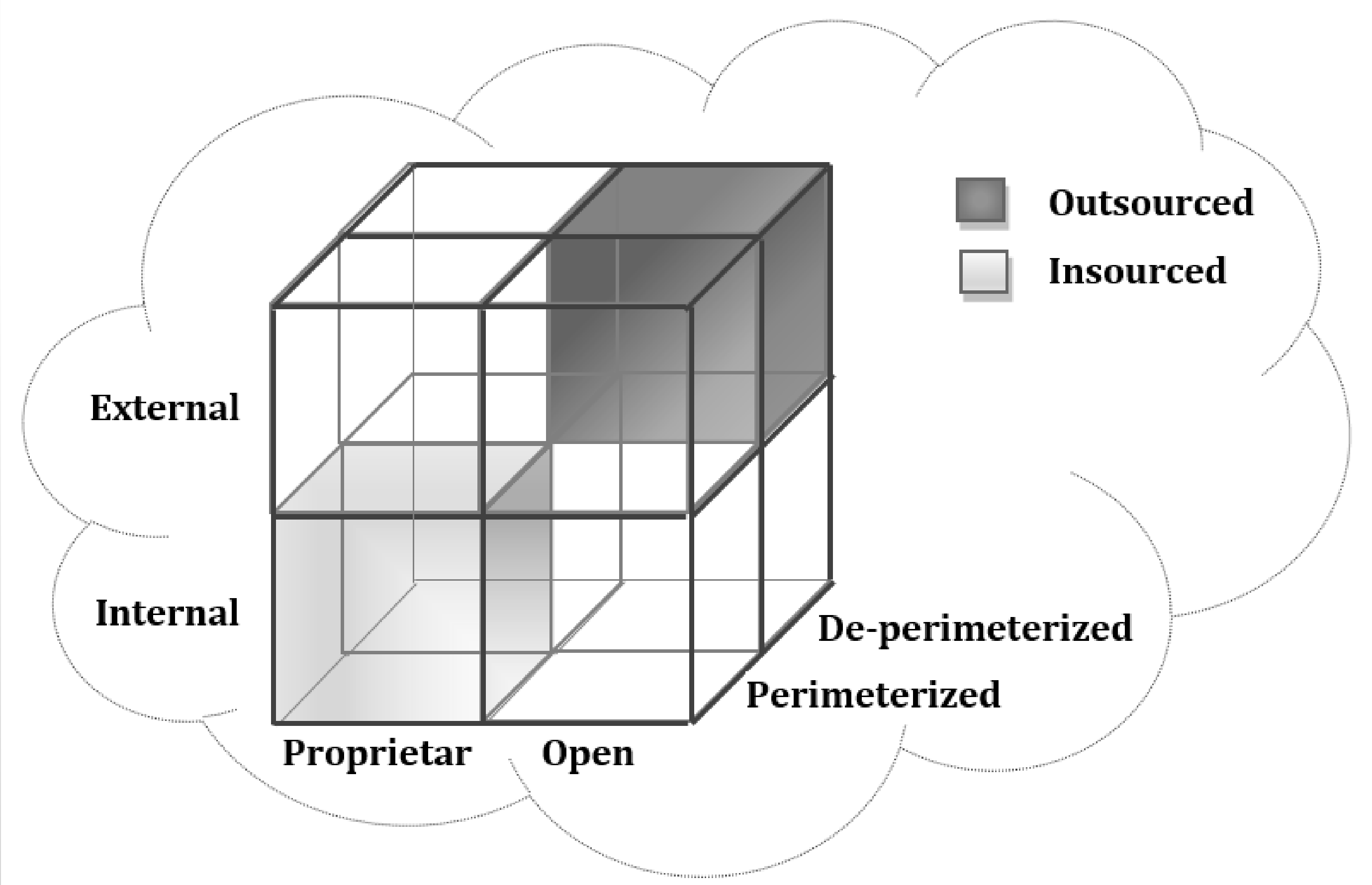
- Sourcing can either be outsourced or insourced for each of the eight cloud forms discussed earlier.

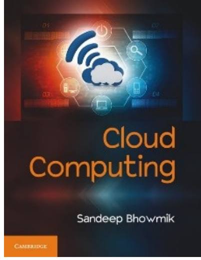
- In Jericho Forum's Cloud Cube Model this fourth dimension is represented by two different colours for painting the cubes.

- Hence the eight smaller cubes that come out after combining the first three dimensions discussed, can take either of the two colours.

The Cloud Cube Model

- Jericho Forum's Cloud Cube Model.

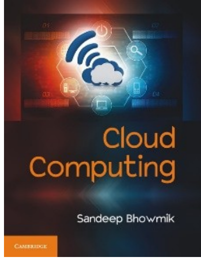




The Cloud Cube Model

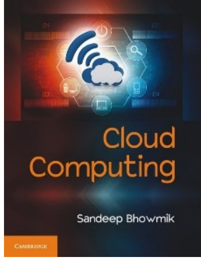
- In this model, the top-right-rear E/O/D-p cloud formation is considered as the "sweet spot" where optimal flexibility and collaboration can be achieved.

- The bottom-left-front I/P/Per cloud formation is the most restricted one.



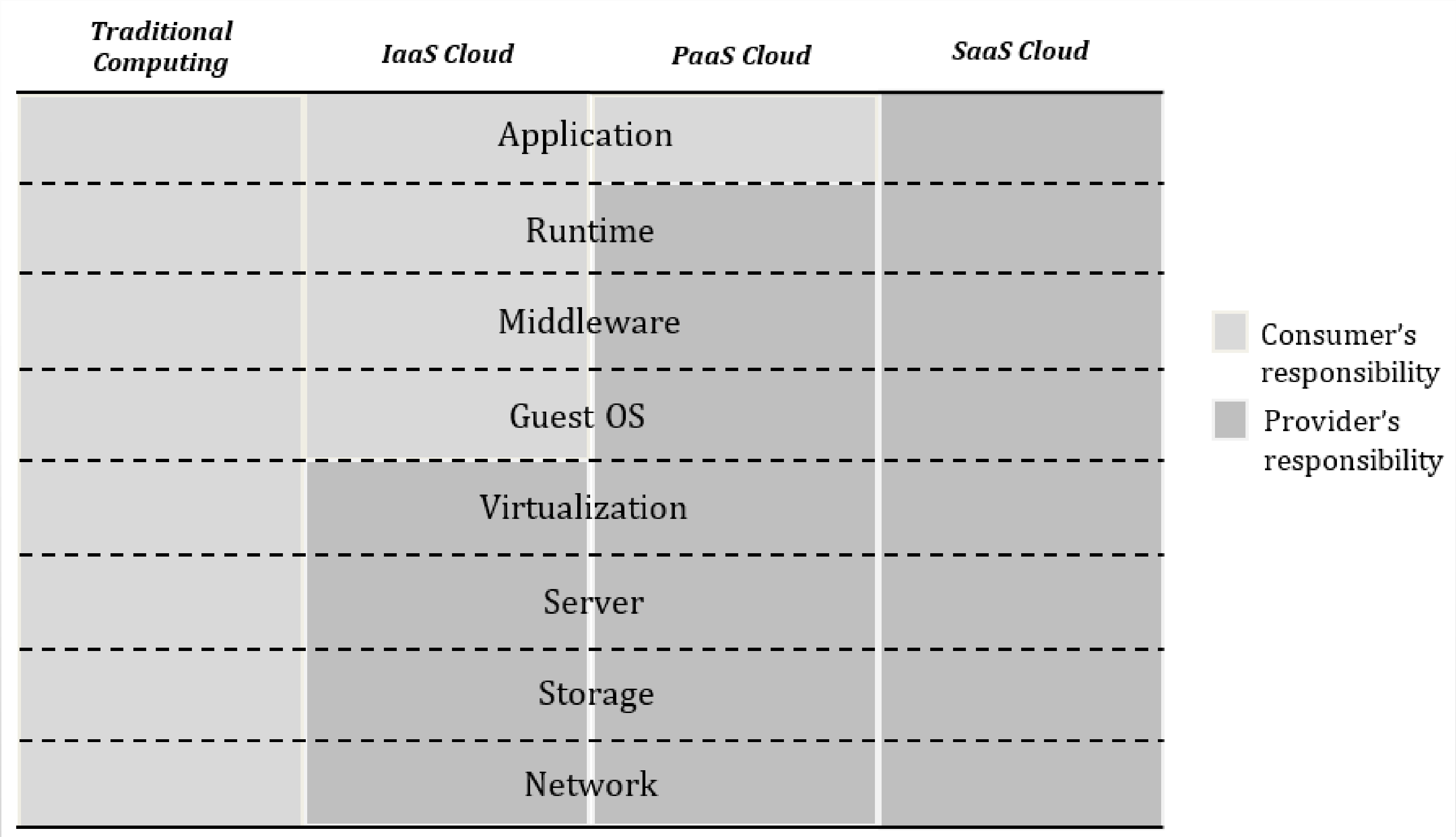
Cloud Security against Traditional Computing

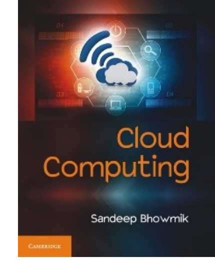
- Collaboration is the tune of cloud based business systems.
- Both consumers and service providers have their share of responsibilities in ensuring adequate security.
- But, unlike traditional computing, consumers no more need to manage everything starting from the bottom of the stack in cloud computing.



Cloud Security against Traditional Computing

- Share of security management responsibilities

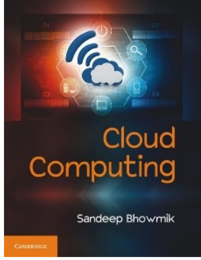




Cloud Security Management

- Security management responsibilities in cloud by service type

Service Type	Security management responsibility at			
	Infrastructure Level	**Platform Level**	**Application Level**	**Operation Level**
Outsourced/IaaS	Provider	Consumer	Consumer	Consumer
Outsourced/PaaS	Provider	Provider	Consumer	Consumer
Outsourced/SaaS	Provider	Provider	Provider	Consumer
Insourced/IaaS	Consumer	Consumer	Consumer	Consumer
Insourced/PaaS	Consumer	Consumer	Consumer	Consumer
Insourced/SaaS	Consumer	Consumer	Consumer	Consumer

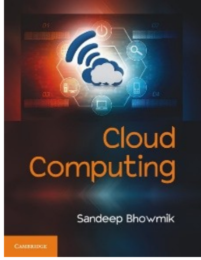


Cloud Security Policy

- Security policies are a set of documentation that guides for reliable security implementation.

- Cloud security strategy define different policies like, system security policies, software policies, and information system policies etc.

- Cloud computing environment also asks organizations to maintain some general policies related to security –
  - Management Policy
  - Regulatory Policy
  - Advisory Policy
  - Informative Policy

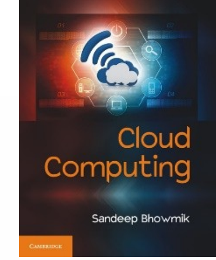


Trusted Cloud Computing

- Trusted computing is a term that refers to technologies, design and policies to develop a highly secure and reliable computing system.

- Trusted cloud computing can be viewed as a way to ensure that the system acts in a predictable manner as intended.

- Reputation or trust building is a time taking process, and larger cloud providers have already taken measures to establish this trust.



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